



Real-Time Thai-to-English Voice Translation Pipeline

Quantization-optimized inference benchmarking for cascaded ASR-NMT-TTS systems

1

Project Goal: Production-Ready Translation Pipeline



Integrated Pipeline

Build a complete Thai-to-English voice translation service combining ASR, NMT, and TTS modules into a single optimized inference pipeline



Real-Time Performance

Achieve sub-second latency with $RTF < 1.0$ for production deployment, enabling conversational voice translation applications



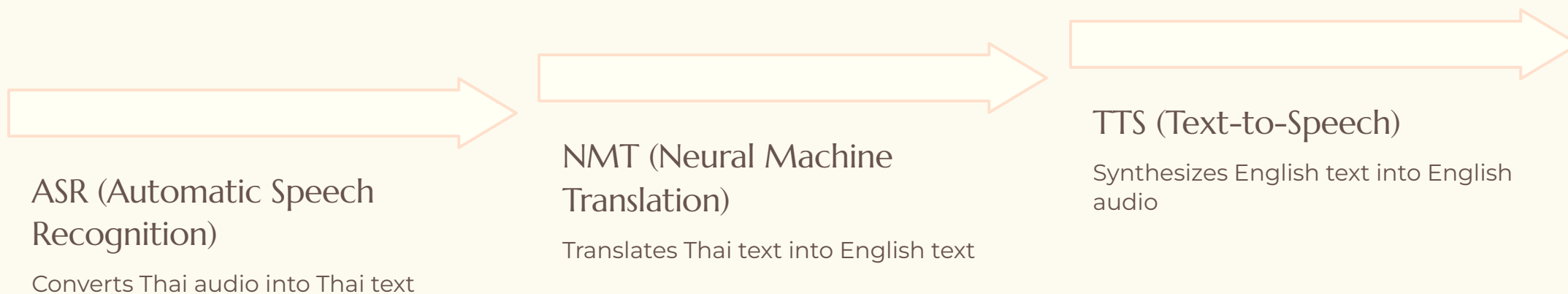
Quantization Optimization

Apply INT8 and FP16 quantization techniques to reduce memory footprint while maintaining accuracy within acceptable degradation thresholds

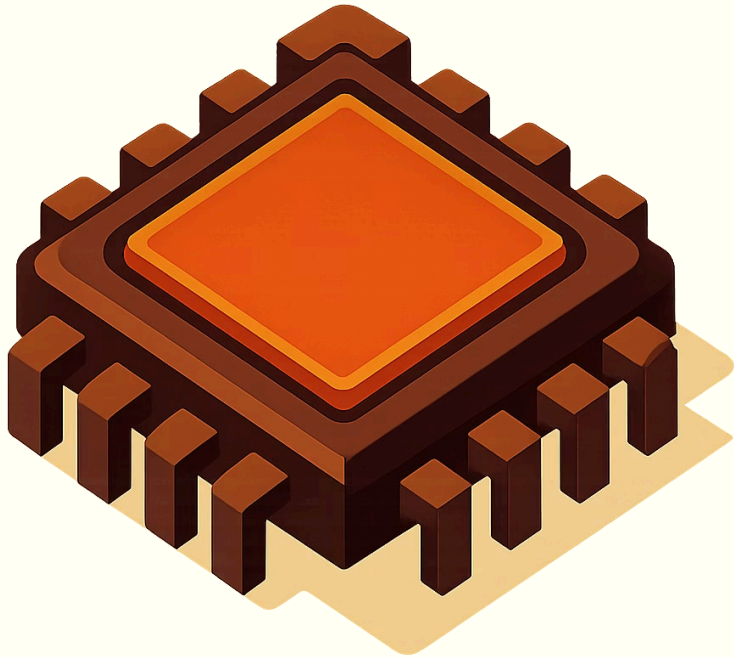
This project bridges research and deployment by transforming pre-trained models into a production-ready service. We leverage open-source datasets and benchmark established methodologies from MT Summit 2025 to validate our cascaded pipeline approach.

Pipeline Architecture

Our cascaded pipeline translates spoken Thai into English in real-time, leveraging a sequence of specialized AI models.



Methodology: Quantization Strategies



1

FP32 Baseline

Full-precision 32-bit floating point models serve as accuracy reference point for all experiments

2

INT8 Dynamic

8-bit integer quantization with dynamic range scaling, reducing model size by 75% with minimal accuracy loss

Each quantization method targets different deployment scenarios: CPU-only inference (INT8), and edge devices with memory constraints (INT8). The choice depends on hardware capabilities and accuracy requirements.

Experimental Setup: Hardware and Datasets

Hardware Configuration

CPU: Standard x86 processors without dedicated AI acceleration for baseline testing

GPU: NVIDIA RTX 5060 Ti 16GB (Blackwell architecture) with CUDA cores for accelerated inference

Tensor Cores: Specialized INT8/FP16 units enabling 2-4x speedup vs FP32 baseline

Dataset Sources

ASR Benchmark: Google FLEURS Thai dataset (1021 sentences) for Thai audio recognition

NMT Benchm: Salesforce WikiText and PyThaiNLP SCB-MT-EnTh-2020 for Thai-English translation

TTS Benchmark: LJSpeech corpus for English text-to-speech synthesis



Thonburian Whisper ASR

Testing Thonburian Whisper Distilled models on Thai audio transcription reveals stark performance improvements. CPU inference shows INT8 significantly outperforming FP32 by dramatically reducing runtime, while GPU deployments leverage Tensor Cores for further speedups.

CPU Summary (FP32 vs INT8)

Model speedups make the CPU pattern immediately clear: INT8 reduces runtime dramatically across all three runs, with progress-003 showing the strongest gain at 9.61x

whisper-th-medium-combined	whisper-th-medium
500 samples	500 samples
FP32 CER 0.205	FP32 CER 0.098
INT8 CER 0.083	INT8 CER 0.096
FP32 RTF-T 11.009	FP32 RTF-T 6.146
INT8 RTF-T 1.248	INT8 RTF-T 0.640

GPU Summary (vs CPU INT8 Baselines)

GPU acceleration makes the contrast even more obvious: each model runs several times faster than its CPU INT8 baseline while keeping CER nearly unchanged.

medium-combined GPU FP16	medium-combined GPU INT8	distill-medium GPU FP16	distill-medium GPU INT8
CPU RTF-T 1.248 → GPU RTF-T 0.210	CPU RTF-T 1.248 → GPU RTF-T 0.267	CPU RTF-T 0.640 → GPU RTF-T 0.070	CPU RTF-T 0.640 → GPU RTF-T 0.080
Speedup 5.95x	Speedup 4.68x	Speedup 9.08x	Speedup 8.00x
CPU CER 0.083 → GPU CER 0.085	CPU CER 0.083 → GPU CER 0.088	CPU CER 0.096 → GPU CER 0.094	CPU CER 0.096 → GPU CER 0.096

Note: RTF-T = Real-Time Factor for transcription. Speedup = FP32 RTF ÷ INT8 RTF.

Note: STT models have not been benchmarked on WER (Word Error Rate) metrics yet. Current results focus on CER (Character Error Rate) and RTF (Real-Time Factor) performance.



REAL-TIME PROCESSING

Typhoon ASR: Real-Time Thai Recognition

Model Architecture

Typhoon-ai/Typhoon-ASR-Realtime using NeMo FastConformer-Transducer architecture

Performance

50× real-time (RTF 0.02) with 84.1% accuracy on FLEURS Thai test set

Quantization

INT8 dynamic quantization shows -0.5% WER change with -4.6% RTF improvement

GPU Acceleration Results

NVIDIA RTX 5060 Ti delivers **71.8% faster inference** vs CPU (RTF 0.006 vs 0.023) while maintaining identical WER. The Transformer encoder benefits from GPU parallelism, but RNN-T autoregressive decoding limits further optimization.

NLLB-600M Translation Benchmark

Model Specifications

Architecture: NLLB-600M parameter model
Task: Thai-to-English machine translation
Dataset: 500 test sentences from FLEURS
Hardware: NVIDIA T4 GPU

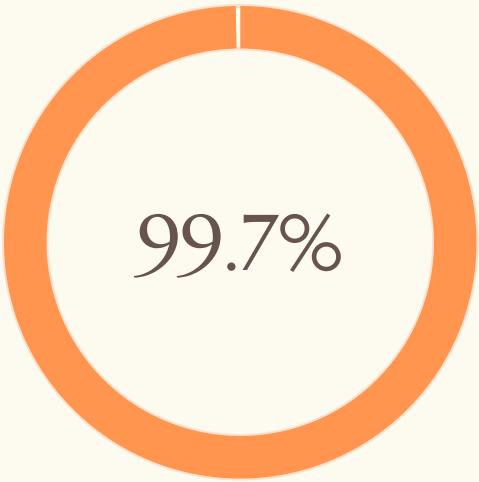
Quantization Results

BLEU Score: 17.87 (FP32) vs 17.81 (INT8)
Latency: 99ms (FP32) vs 54ms (INT8)
Throughput: 13K WPM (FP32) vs 24K WPM (INT8)
Accuracy Drop: Only -0.3% BLEU



Words Per Minute

INT8 quantized throughput on T4 GPU



Accuracy Retention

BLEU score preserved vs FP32 baseline

GPT-4.1 nano Translation Benchmark

◆ Model Specifications

Model	GPT-4.1 nano
Architecture	Large Language Model (LLM API)
Task	Thai-to-English translation
Dataset	500 test sentences (SCB MT)
Hardware	API-based (no local GPU required)

◆ Performance Results

BLEU Score	23.18
Latency	0.75 sec / request
Throughput	1765 WPM
Total Cost	\$0.0098 (500 samples)

◆ Efficiency Metrics

Cost per sentence	
\$0.000020	

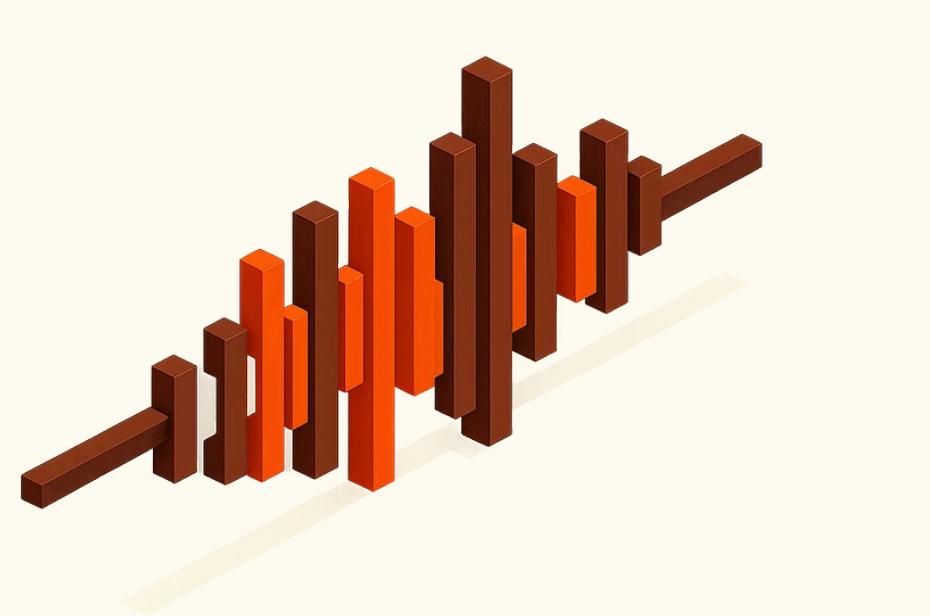
NLLB-600M vs GPT-4.1 nano Comparison

NLLB-600M	GPT-4.1 nano
BLEU: 17.8	BLEU: 23.18
Latency: 54 ms	Latency: 750 ms
Throughput: 24K WPM	Throughput: 1.7K WPM
Cost: Free	Cost: \$0.000020 per sentence

📌 NLLB excels in speed and efficiency, while GPT-4.1 nano provides better semantic quality at a minimal cost.

TTS Engine Comparison: Piper vs Kokoro

Two text-to-speech engines demonstrate trade-offs between speed and quality. Piper achieves real-time performance (RTF 0.19) with moderate accuracy, while Kokoro delivers higher fidelity at slower speeds (RTF 1.46).



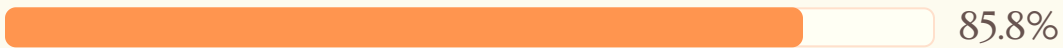
Piper (ONNX)

RTF: 0.186 (5.4× real-time)

RAM: 37.55 MB

Quality: WER 14.2%, PESQ 1.05

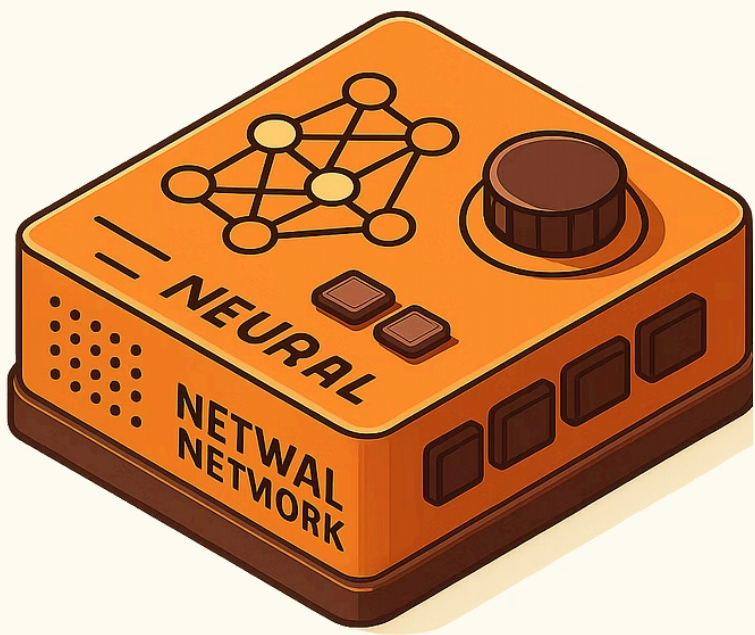
Best For: Low-latency applications



Piper Accuracy

Word error rate of 14.2% across 100 sentences

Note: TTS models have not been benchmarked on GPU yet. Current results are CPU-only performance metrics.



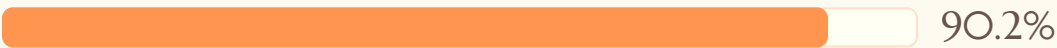
Kokoro-82M

RTF: 1.46 (0.69× real-time)

RAM: 288.62 MB

Quality: WER 9.8%, PESQ 1.10

Best For: High-fidelity output



Kokoro Accuracy

Lower WER of 9.8% demonstrates superior intelligibility



Pipeline Integration Challenges & Future Outlook

As we move towards a production-ready pipeline, several key challenges and strategic decisions must be addressed to ensure optimal performance and deployment.

Current Integration Challenges

- **Decentralized Codebase:** The current pipeline components are fragmented, running independently on individual team members' machines.
- **Inconsistent Measurement:** This dispersed setup hinders accurate and consistent end-to-end performance measurement.

Deployment Strategy & Future Decisions

- **CPU vs. GPU Performance:** Initial benchmarks confirm that CPU-only inference, while ideal for local/mobile, struggles with real-time speed requirements.
- **GPU Transition:** We must transition to GPU acceleration to meet latency targets.
- **Deployment Dilemma:** The next step is to decide whether to develop a local GPU demonstration for use case validation or proceed directly to server-side deployment.



Next Steps: Pipeline Integration Roadmap

01

Module Integration

Combine Whisper ASR, NLLB-NMT, and Kokoro-TTS into single inference pipeline with VAD-based chunking

02

End-to-End Optimization

Apply CTranslate2 or ONNX runtime for cross-model graph optimization and reduced memory copies

03

Production Deployment

Develop fully functional real use case of real-time audio translation

Reference Implementations

- **Research Paper:** "Speech-to-Speech Translation Pipelines for Conversations in Low-Resource Languages" (MT Summit 2025) - Demonstrates cascaded pipeline approach with independent module optimization
- **GitHub:** Hugging Face speech-to-speech - Modular pipeline with VAD, STT, MT, and TTS components for local deployment
- **Key Insight:** Performance ranks of individual modules are independent, allowing selection of best-in-class ASR + best-in-class MT + best-in-class TTS

Any Questions?

We'd love to hear your thoughts.

Pipeline Architecture

Our cascaded pipeline translates spoken Thai into English in real-time, leveraging a sequence of specialized AI models.



Thai Audio Input

Raw Thai speech captured from the user.



ASR (Thai)

Converts Thai audio into written Thai text.



Thai Text

Transcribed text in the source language.



NMT (Thai-English)

Translates Thai text into English text.



English Text

Translated text in the target language.



TTS (English)

Synthesizes spoken English audio from text.



English Audio Output

Translated and synthesized speech delivered to the user.

Quantization Impact: The Hardware Divide



1

CPU Performance Paradox

Standard x86 processors lack native 8-bit math support, forcing constant INT8-to-FP32 conversion during inference. This dequantization overhead makes INT8 *slower* than FP32 despite smaller data size.

Result: INT8 reduces RAM usage by 50% but provides no speed benefit on CPU-only systems

2

GPU Acceleration Advantage

Modern GPUs contain specialized Tensor Cores designed for INT8/FP16 operations. These skip conversion overhead entirely, processing 8-bit data natively at 2-4x FP32 throughput.

Result: INT8 delivers 50% memory savings plus 2-4x speedup on GPU hardware



Key Insight: Hardware determines quantization strategy. CPU deployments prioritize memory savings over speed. GPU deployments achieve both reduced memory and accelerated inference.

Thonburian Whisper ASR

Testing Thonburian Whisper Distilled models on Thai audio transcription reveals stark hardware dependencies. CPU inference shows FP32 outperforming INT8 due to dequantization overhead, while GPU leverages Tensor Cores for dramatic speedups.

CPU Summary (FP32 vs INT8)

Model	Samples	FP32 CER	INT8 CER	FP32 RTF-T	INT8 RTF-T	Speedup
progress-001 (medium-combined, 100)	100	0.103	0.100	10.856	1.241	8.75x
progress-002 (medium-combined, full)	500	0.205	0.083	11.009	1.248	8.82x
progress-003 (distill-medium, full)	500	0.098	0.096	6.146	0.640	9.61x

GPU Summary (vs CPU INT8 Baselines)

Model	CPU RTF-T	GPU RTF-T	Speedup	CPU CER	GPU CER
medium-combined GPU FP16 (run_009)	1.248	0.210	5.95x	0.083	0.085
medium-combined GPU INT8 (run_010)	1.248	0.267	4.68x	0.083	0.088
distill-medium GPU FP16 (run_011)	0.640	0.070	9.08x	0.096	0.094
distill-medium GPU INT8 (run_012)	0.640	0.080	8.00x	0.096	0.096

Note: RTF-T = Real-Time Factor for transcription. Speedup = FP32 RTF ÷ INT8 RTF.

Note: STT models have not been benchmarked on WER (Word Error Rate) metrics yet. Current results focus on CER (Character Error Rate) and RTF (Real-Time Factor) performance.